

Independent Content and Style Evidence for UAV–Satellite Retrieval

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Local working draft: completed-result integration; remaining experiments and final review are pending.

Abstract—Fixed semantic vocabularies offer a compact way to compare content and acquisition-style information in UAV–satellite retrieval. We study an interface with independent per-image encoding: a frozen vision–language model scores each image against both vocabularies and separate probability encoders fuse the resulting vectors with a learned visual descriptor. A common multi-positive training protocol evaluates six input configurations on University-1652 and SUES-200. Across the 11 official retrieval tasks, the Full configuration has lower three-seed mean Recall@1 and official trapezoidal mAP than Visual on 10 tasks; the Street2S exception has low absolute accuracy. In bidirectional cross-dataset transfer, Full has lower mean Recall@1 in all 11 target tasks. Seed-1 corruption measurements and clean-query margin diagnostics characterize these results under explicit metric and membership definitions. A separate historical study of four public visual baselines analyzes ranking and score fusion. The completed comparisons do not support a general accuracy advantage from this frozen semantic-fusion design and delimit the claims that its conditional diagnostics can support.

Index Terms—Cross-view geo-localization, multimodal retrieval, semantic evidence, unmanned aerial vehicle localization, vision–language models.

I. INTRODUCTION

Cross-view geo-localization retrieves a geo-referenced image of the place observed by a query camera [1]. For an unmanned aerial vehicle (UAV), satellite imagery provides a reference gallery across locations without a ground-level survey of every route. A UAV image can query this gallery to identify its scene, and a satellite image can retrieve the corresponding observations in a UAV image collection. University-1652 and SUES-200 organize these two directions into identity-based retrieval tasks [2], [3]. The retrieved scene provides a location hypothesis for subsequent geometric localization.

Recognizing the same place across views requires more than matching local appearance. An oblique UAV view exposes building facades and nearby structures, whereas a satellite view emphasizes roofs, roads, and the surrounding layout. Flight altitude changes the visible footprint and the relative scale of these structures. Shadows, vegetation, and sensor response add further appearance variation. At the same time, different locations may contain similar buildings, parking areas, or road patterns, as illustrated in Fig. 1. A useful descriptor must preserve the scene information that distinguishes these locations across the view change.

Visual approaches address this problem through descriptor learning and spatial aggregation. Global encoders summarize the image in a common embedding space [4], [5]. LPN, FSRA,

CA-HRS, and SDPL retain local patterns or align feature regions to represent finer scene structure [6]–[9]. Contrastive training directly increases the similarity of matching locations while separating other locations [10]. These methods primarily organize visual features. Image–text representations provide another source of information: a score for a concept such as a road intersection or dense building layout can describe an image through a vocabulary shared by both views.

Recent multimodal methods integrate this information at different stages. CDM-Net uses remote-sensing descriptions in a view-synthesis and structural-matching pipeline [11]. VLGeo builds attribute-based descriptions from image triplets and learns cross-view, cross-modal alignment [12]. These approaches combine semantic input construction with a dedicated interaction mechanism. We consider a simpler input interface: a fixed semantic vector computed separately for each image. This interface allows the reference gallery to be encoded before a query arrives and exposes the contribution of each semantic input through branch-level comparisons.

The design distinguishes geographic content from acquisition style. Content describes visible scene elements, including buildings, roads, and vegetation. Style describes viewpoint and appearance conditions, including altitude, illumination, and contrast. Grouping these concepts separately preserves their different roles when the scores enter the retrieval model. Content scores can associate visually different observations with common scene concepts; style scores provide information about the conditions under which those concepts appear. We use these groups as explicit input categories and learn their relation to visual features in the fusion head.

Our framework scores each image against two fixed vocabularies using a frozen vision–language model. A trainable ResNet produces the visual descriptor, and two independent MLPs encode the content and style scores. The normalized branch descriptors are concatenated and projected to a common retrieval space. The query and gallery images share encoder parameters; each descriptor depends on its own image and the fixed texts. A symmetric multi-positive contrastive objective accounts for the multiple UAV observations associated with one location.

Evaluation must also distinguish representation quality from the effect of the retrieval procedure. Local score normalization can change the leading match, query expansion can reorder lower-ranked positives, and a smaller gallery removes difficult competitors. We study these effects with four public visual baselines and retain the same feature sources within each comparison. The resulting analysis motivates explicit controls for ranking, gallery composition, and model combination in semantic-source comparisons. The contributions of this work

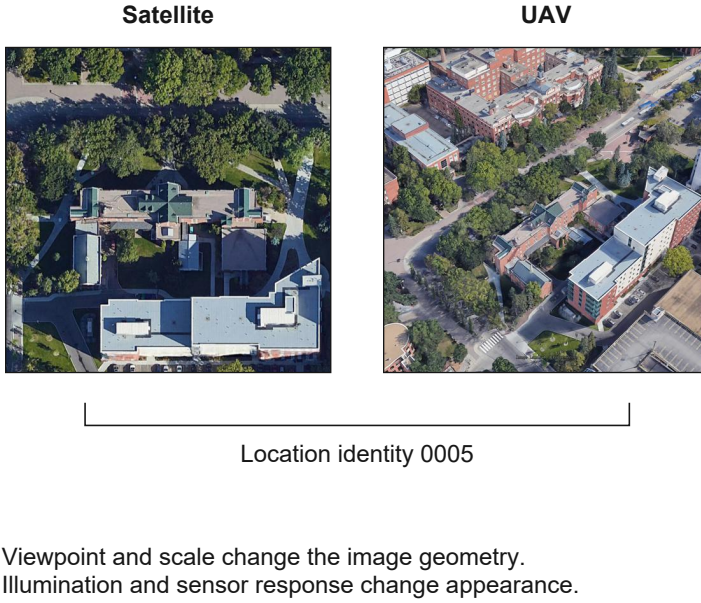
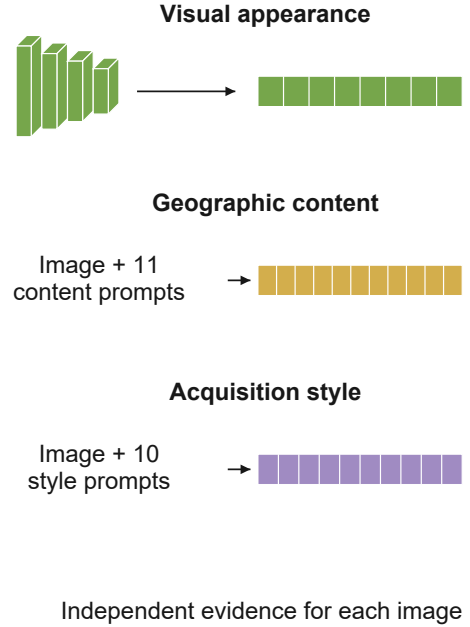
(a) The same place from two views**(b) Image evidence**

Fig. 1. Cross-view observations and image evidence. (a) Satellite and UAV images from location 0005 in University-1652 show changes in geometry and appearance across viewpoints. (b) The visual branch describes image appearance, while frozen CLIP scores each image against 11 content prompts and 10 acquisition-style prompts. The vector glyphs depict the representation structure.

are summarized as follows:

- We define fixed content and acquisition-style input vectors that are constructed separately for each image and can be cached for independent query and gallery encoding.
- We evaluate six input configurations under one multi-positive training protocol, retaining all official tasks and all three seeds. The Full-versus-Visual comparison is negative on 10 of 11 official tasks for both mean R@1 and mAP, and on all 11 transfer tasks for mean R@1.
- We distinguish seed-1 corruption retention from absolute accuracy, and shared-query margin strata from variant-specific selective retrieval. These diagnostics define the comparison being made without assigning a causal explanation or a calibrated-probability interpretation.
- We retain a separate analysis of four public visual baselines on University-1652, covering ranking rules, gallery size and retrospective score fusion under their own trained feature sources.

Section II reviews related approaches. Sections III and IV describe the architecture and experimental protocol. Section V reports the semantic-model results, followed by the separate historical public-baseline analysis in Section VI.

II. RELATED WORK

A. Visual Representations for Cross-View Retrieval

Cross-view retrieval methods preserve scene information through different feature organizations. CVM-Net learns a

common embedding for ground and aerial imagery [4]. Spatial aggregation and Transformer-based encoders incorporate structure and context into the descriptor [5], [13]. For UAV–satellite retrieval, LPN pools concentric local regions, FSRA aligns segmented feature regions, CA-HRS selects hierarchical representations, and SDPL aggregates shifted dense partitions [6]–[9]. Their different spatial units address the displacement, scale, and appearance changes between views.

CAMP combines contrastive attribute mining with position-aware partitioning [14]. DAC combines a domain space alignment module for fine-grained features with cross-batch scene consistency [15].

Training objectives provide complementary constraints. Sample4Geo combines symmetric contrastive learning with negative sampling [10]. Classifier interaction, causal representation learning, multilevel alignment, and query-driven feature learning further shape the cross-view embedding [16]–[19]. DSTG uses multiscale logit-standardization distillation to transfer a Swin Transformer teacher to a smaller student [20]. Its ground-to-satellite experiments on CVUSA and CVACT address model efficiency in a different view setting. Our visual branch uses a ResNet with global pooling, which provides a common visual input for the content and style branch comparisons.

Recent preprints extend this line of work toward efficiency, geometry, and cross-dataset transfer. MobileGeo studies hierarchical knowledge distillation for compact retrieval encoders [21]. MGS2-Net incorporates a depth prior into geometric representation learning [22], while InfoGeo stud-

ies information-theoretic, object-centric learning for transfer across geo-localization datasets [23]. Their encoder inputs and evaluation settings are retained in the reference comparisons below.

B. Semantic Inputs and Multimodal Matching

CLIP learns a common space for image and text embeddings [24]. In geo-localization, textual information can describe a query, condition an image-generation stage, or guide feature alignment. GeoText-1652 provides natural-language descriptions and spatial-relation matching [25]. CDM-Net combines perspective transformation, remote-sensing descriptions, and conditional diffusion before matching point–line structure in the synthesized view [11]. The descriptions therefore contribute to both the synthesized image and its structural representation.

VLGeo generates attribute prompts from triplet images and learns bidirectional-guided semantic alignment with contrastive supervision [12]. Its ablations examine modal combinations, description attributes, language models, and alignment components. We instead use the image–text encoder to score fixed concepts separately for each image. Content and style probabilities enter two learned branches alongside the visual descriptor. The common vocabulary gives every image the same semantic coordinates and supports independent gallery encoding. This compact representation captures concept scores; detailed descriptions and explicit spatial relations provide a richer, different input interface.

C. Environmental Variation and Localization Settings

Environmental changes affect both image features and semantic scores. MuSe-Net models environmental style in aerial retrieval [26]. EAGLe combines a dual-objective discriminator with a style-mixture module: adversarial training promotes view and environment invariance, while feature-statistic mixing broadens the styles encountered during training [27]. Our style branch follows a conditioning approach. It retains the image’s acquisition-style scores as auxiliary input to descriptor fusion. Clean and corrupted-query comparisons measure how that conditioning changes retrieval under appearance shifts.

Other geo-localization settings produce different outputs or consume additional observations. Vector-map retrieval uses map geometry, region-to-point localization links regional and point-level matches, and iterative rendering combines multiple UAV observations [28]–[30]. Game4Loc evaluates continuous-region and non-centered localization [31]. The present study uses ranked whole-image identity retrieval on University-1652 and SUES-200. We specify gallery composition, query view, and ranking rules for each comparison because these settings determine the candidate matches available to a descriptor.

III. INDEPENDENT SEMANTIC EVIDENCE FUSION

A. Task Formulation and Framework Overview

Let $\mathcal{X}^q = \{(x_i^q, y_i)\}$ and $\mathcal{X}^g = \{(x_j^g, y_j)\}$ denote query and gallery images, where y identifies a location. The query and gallery roles are assigned by the evaluation task; they do not

change the image encoder. For active branches $\mathcal{A} \subseteq \{v, c, s\}$, the descriptor is

$$\mathbf{z} = F_\theta(x, \mathbf{p}^c(x), \mathbf{p}^s(x); \mathcal{A}) \in \mathbb{R}^d, \quad (1)$$

where only inputs selected by $\mathcal{A}$ are consumed. Superscripts v , c , and s denote visual, content, and style branches. The same parameters θ encode the two views independently. We use the stable normalization

$$\mathcal{N}(\mathbf{u}) = \frac{\mathbf{u}}{\max(\|\mathbf{u}\|_2, \varepsilon)}, \quad \varepsilon = 10^{-12}. \quad (2)$$

The output is L_2 -normalized and retrieval sorts cosine scores:

$$s_{ij} = \mathbf{z}_i^{q\top} \mathbf{z}_j^g, \quad \pi_i = \text{argsort}_j^\downarrow s_{ij}. \quad (3)$$

Here π_i lists gallery indices in descending similarity order.

Figure 2 separates offline evidence construction from trainable retrieval. Frozen CLIP encoders create two probability vectors for each image. A ResNet branch encodes visual appearance, while two independent multilayer perceptrons (MLPs) encode content and style probabilities. A multimodal variant concatenates its normalized branch descriptors and learns a fusion projection; a single-branch variant returns that branch directly. Location labels determine positive matches in the training objective, but are not encoder inputs. At inference, each image is encoded once before query–gallery similarities are computed.

B. Fixed Content and Style Evidence

We separate what an image contains from how it is acquired so that their retrieval effects can be tested separately. The content vocabulary $\mathcal{C} = \{c_k\}_{k=1}^{11}$ covers campus layout, buildings, roads, intersections, parking, vegetation, sports fields, open areas, water, dense layout, and stable spatial layout. The style vocabulary $\mathcal{S} = \{s_l\}_{l=1}^{10}$ covers UAV/satellite viewpoint, altitude, shadows, illumination, contrast, sensor gap, and seasonal vegetation. The two groups define the prompt categories used by the probability branches. Their scores can be correlated across images.

Frozen OpenAI CLIP ViT-B/32 image and text encoders run in FP16 [24]. For image embedding $e(x)$ and text embedding $t(w)$, normalization gives

$$\hat{e}(x) = \frac{e(x)}{\|e(x)\|_2}, \quad \hat{t}(w) = \frac{t(w)}{\|t(w)\|_2}. \quad (4)$$

Each vocabulary is scored separately:

$$p_k^c(x) = \frac{\exp(\tau \hat{e}(x)^\top \hat{t}(c_k))}{\sum_{r=1}^{11} \exp(\tau \hat{e}(x)^\top \hat{t}(c_r))}, \quad (5)$$

$$p_l^s(x) = \frac{\exp(\tau \hat{e}(x)^\top \hat{t}(s_l))}{\sum_{r=1}^{10} \exp(\tau \hat{e}(x)^\top \hat{t}(s_r))}. \quad (6)$$

The frozen scale is $\tau = 100$, and the text order is fixed. These are relative scores normalized within each vocabulary. The vocabulary and its ordering therefore remain fixed throughout a comparison.

The cache stores $\mathbf{p}^c \in \mathbb{R}^{11}$ and $\mathbf{p}^s \in \mathbb{R}^{10}$ under each relative image path. CLIP receives the image pixels and the fixed texts. Training labels are joined by path when constructing positive

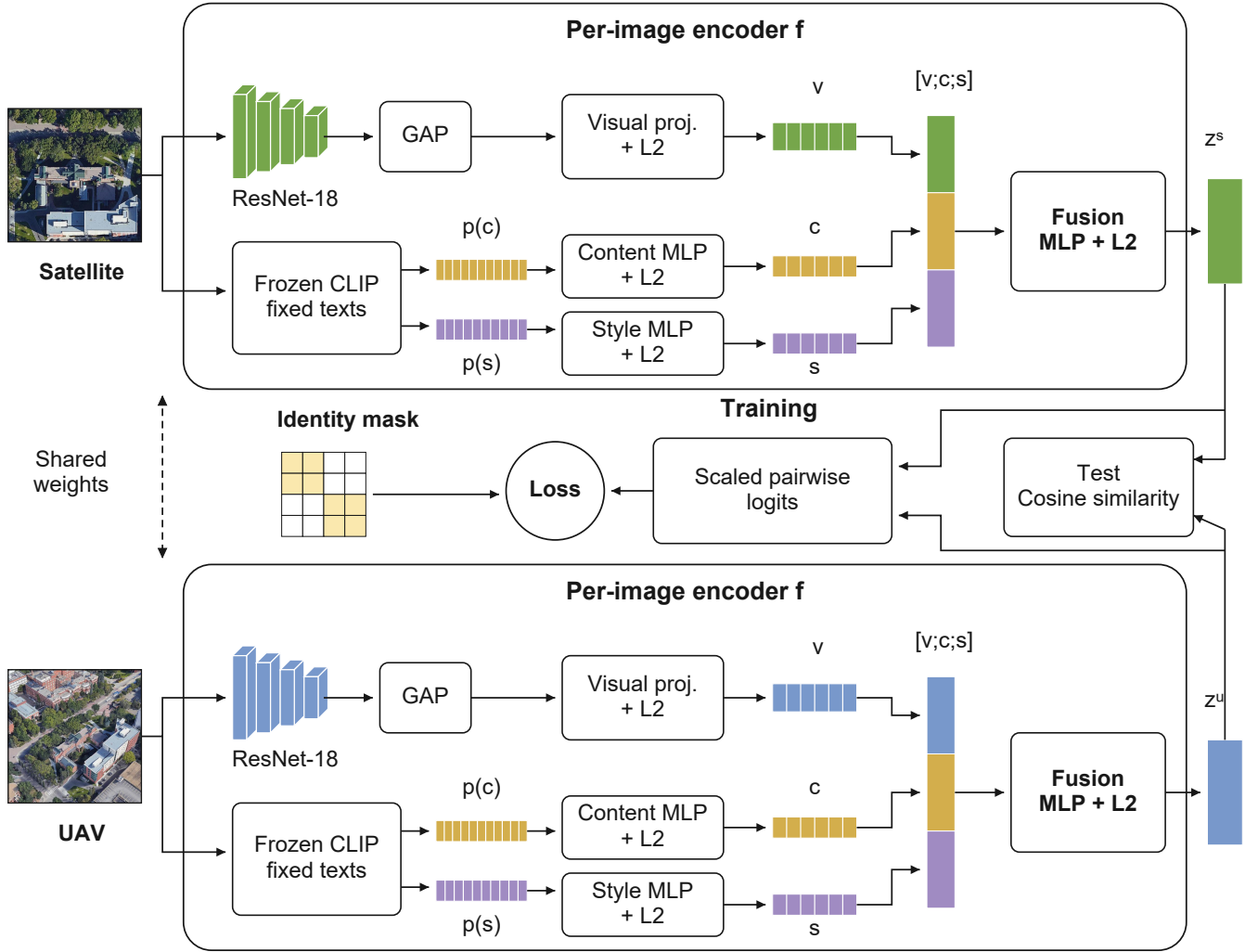


Fig. 2. Overview of the full visual-content-style retrieval model. Satellite and UAV images follow two independent calls to the same per-image encoder f , with shared trainable parameters. ResNet-18 followed by global average pooling and a visual projection produces v . Frozen CLIP computes separate softmax probability vectors over the fixed content and style vocabularies, which separate MLPs project to c and s . Each branch is L_2 -normalized. The concatenation $[v; c; s]$ is projected and normalized to a 512-dimensional descriptor. During training, the two descriptor batches form scaled pairwise logits and a same-identity mask defines the symmetric multi-positive objective. At test time, cosine similarity ranks independently encoded gallery descriptors. Green and blue denote satellite and UAV visual features, amber denotes content, and lavender denotes style.

pairs for the retrieval objective. Content-only and style-only variants use the corresponding probability vector; visual-plus-one-branch variants combine that vector with learned visual features.

C. Visual and Probability Encoders

The visual branch retains a learned image representation beyond the fixed semantic scores. An ImageNet-initialized ResNet-18 or ResNet-50 [32] produces a globally pooled feature $g = f_{\text{cnn}}(x) \in \mathbb{R}^C$, where $C = 512$ or $C = 2048$, respectively. The final classifier is removed, and the globally pooled final feature map supplies the visual projection. Let $A_r^a(u) = W_r^a u + b_r^a$ be an affine layer, LN_n layer normalization over n entries, σ the Gaussian error linear unit (GELU),

and D_ρ dropout with probability ρ . The visual projection is

$$h^v = D_\rho[\sigma(\text{LN}_d(A_1^v(g)))] , \quad (7)$$

$$z^v = \mathcal{N}(A_2^v(h^v)) . \quad (8)$$

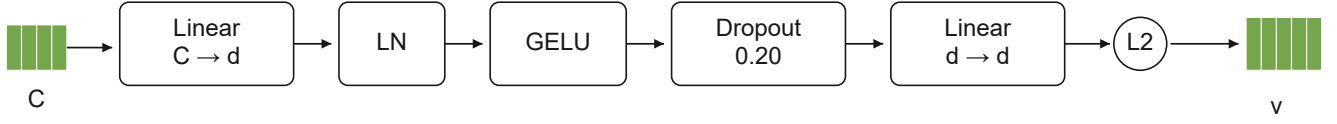
The affine widths are $C \rightarrow d \rightarrow d$. Dropout is active during training and disabled during evaluation.

The probability branches adapt the low-dimensional fixed scores to the retrieval space without updating CLIP. For $a \in \{c, s\}$, let $n_c = 11$ and $n_s = 10$. Each branch has its own parameters and follows

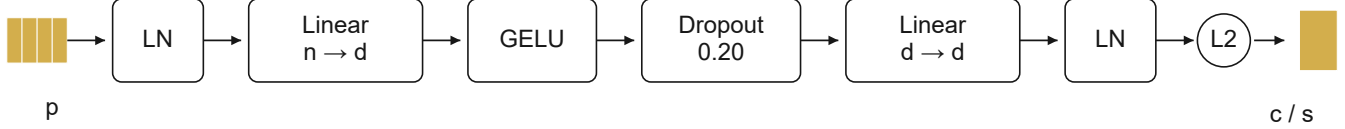
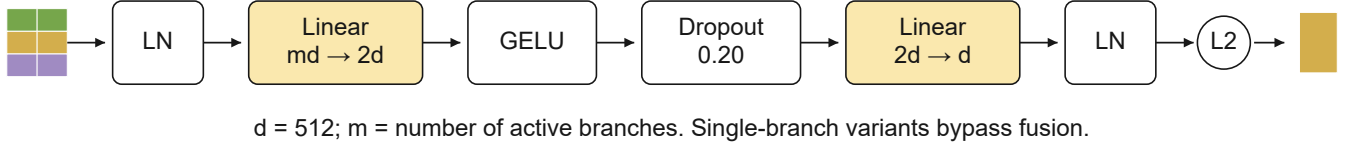
$$h^a = D_\rho[\sigma(A_1^a(\text{LN}_{n_a}(p^a)))] , \quad (9)$$

$$z^a = \mathcal{N}[\text{LN}_d(A_2^a(h^a))] . \quad (10)$$

The widths are $n_a \rightarrow d \rightarrow d$. Unlike the visual projection, a probability MLP normalizes its input before its first affine layer and applies another layer normalization after its second affine layer. Figure 3 summarizes the normalized branch

(a) Visual projection**(b) Content and style projections**

Two independently parameterized MLPs: $n = 11$ for content, $n = 10$ for style

**(c) Descriptor fusion****(d) Branch configurations**

	V	C	S
V	■	□	□
C	□	■	□
S	□	□	■
V+C	■	■	□
V+S	■	□	■
V+C+S	■	■	■

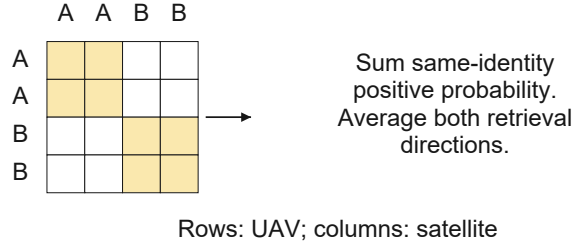
(e) Multi-positive objective

Fig. 3. Projection layers, branch configurations, and the training objective. (a) The visual projection takes C -dimensional pooled features, where $C = 512$ for ResNet-18 and $C = 2048$ for the ResNet-50 sensitivity model. (b) Content and style use distinct MLP parameters with input dimensions $n = 11$ and $n = 10$, respectively. (c) For m active branches, the fusion head maps md to $2d$ and then d , with $d = 512$ in the main configuration; a single active branch bypasses this head. LN denotes layer normalization, and L_2 denotes normalization of the output vector. (d) The six input configurations. (e) An illustrative four-pair identity mask, in which every same-identity pair contributes to the positive probability mass. The objective averages both retrieval directions.

descriptors, their controlled fusion, and the positive-mask construction. Each probability branch processes one image, and the fusion head learns how its scores interact with the visual descriptor.

D. Branch Configuration and Descriptor Fusion

Keeping separate encoders makes the source comparisons explicit, but their outputs must still be combined for joint retrieval. Table I defines the active branch set for each variant. The branch configuration is fixed before training. For a multimodal variant with $m = |\mathcal{A}|$, the concatenation follows the order visual, content, style with inactive branches omitted:

$$\mathbf{h}_{\mathcal{A}} = \mathbf{z}^{a_1} \parallel \dots \parallel \mathbf{z}^{a_m} \in \mathbb{R}^{md}. \quad (11)$$

The fusion head is

$$\mathbf{r} = D_{\rho} \left[\sigma \left(A_1^f (\text{LN}_{md}(\mathbf{h}_{\mathcal{A}})) \right) \right], \quad (12)$$

$$\mathbf{z} = \mathcal{N} \left[\text{LN}_d(A_2^f(\mathbf{r})) \right], \quad (13)$$

with widths $md \rightarrow 2d \rightarrow d$. For $m = 1$, the fusion head is absent and $\mathbf{z} = \mathbf{z}^{a_1}$. Thus every variant has the same output dimension, while multimodal variants learn how their active sources interact.

The primary contrasts are visual+content versus visual, visual+style versus visual, and full versus each two-branch variant. These comparisons test the effect of adding evidence under a common training protocol. The output width is shared,

TABLE I
INPUT CONFIGURATIONS. ALL ACTIVE ENCODERS ARE LEARNED UNDER THE SAME RETRIEVAL OBJECTIVE; THE CLIP EVIDENCE REMAINS FROZEN.

Variant	Visual	Content	Style
visual	✓		
content		✓	
style			✓
visual+content	✓	✓	
visual+style	✓		✓
full	✓	✓	✓

while the added probability encoders and fusion layers increase the parameter count.

E. Multi-Positive Contrastive Training

Several UAV observations can correspond to one location, so treating only the diagonal pair as positive would penalize valid same-location matches. Following the contrastive retrieval formulation [10], we therefore optimize the probability mass assigned to all positives. A batch contains $B_I = 16$ identities and $B_K = 4$ query instances per identity, giving $B = B_I B_K$. Each query receives a same-identity satellite image selected deterministically from its path, epoch, and seed. For trainable log-scale α , the logits and positive mask are

$$L_{ij} = \gamma \mathbf{z}_i^{q^\top} \mathbf{z}_j^g, \quad M_{ij} = \mathbb{1}[y_i = y_j], \quad (14)$$

$$\gamma = \exp[\text{clamp}(\alpha, \log 5, \log 100)]. \quad (15)$$

The scale is initialized with $\alpha = \log(1/0.07)$ and lies in $[5, 100]$ during the forward pass. The directional objective is

$$\mathcal{L}_{q \rightarrow g} = \frac{1}{B} \sum_i \left[\log \sum_j e^{L_{ij}} - \log \sum_{j: M_{ij}=1} e^{L_{ij}} \right]. \quad (16)$$

Both sums are evaluated with log-sum-exp; the batch is rejected if an anchor has no positive. The training objective is

$$\mathcal{L} = \frac{1}{2}(\mathcal{L}_{q \rightarrow g} + \mathcal{L}_{g \rightarrow q}), \quad (17)$$

where the reverse direction uses the transposed logits and mask. The same objective is used for every input configuration, holding the supervision rule fixed across branch comparisons.

IV. EVALUATION SETTINGS

The protocol compares content and style inputs under the same image encoder, training schedule, and retrieval tasks. Section VI reports a complementary analysis of four public visual baselines with fixed feature sources. The completed semantic-model evidence comprises 42 fits and 231 official run-task evaluations, followed by transfer, seed-1 corruption and clean-query diagnostics. Later external-comparison and efficiency results are not represented as completed experiments here. Author-checkpoint re-evaluation and independent training for CAMP and DAC remain unfinished. Their published results are distinct from the completed project evaluations.

A. Datasets and Retrieval Tasks

University-1652. The semantic-model protocol uses the 701 official training identities, with 40,513 drone/street training queries and 701 same-identity satellite positives. The identity-disjoint test defines drone-to-satellite (D2S), satellite-to-drone (S2D), and street-to-satellite (Street2S) retrieval [2]. D2S uses 37,855 queries from 701 identities against 951 satellite images; S2D uses 701 queries against 51,355 drone images; Street2S uses 2,579 queries against 951 satellites. Every gallery retains 701 target identities and 250 distractor identities. The completed public-baseline experiments concern D2S and S2D only.

SUES-200. The fixed manifest assigns 120 identities to training and 80 to testing [3]. A semantic-model fit uses 24,000 UAV images from four heights and 120 training satellite images. Separate retrieval tasks are defined at 150, 200, 250, and 300 m in both directions. At each height, UAV-to-satellite retrieval has 4,000 held-out UAV queries and 200 satellite candidates; satellite-to-UAV retrieval has 80 held-out satellite queries and 10,000 UAV candidates. The full galleries retain all 200 identities, including the 120 training identities as distractors. Results are specified per height and direction rather than combined with a duplicated overall score. Figure 4 summarizes both datasets.

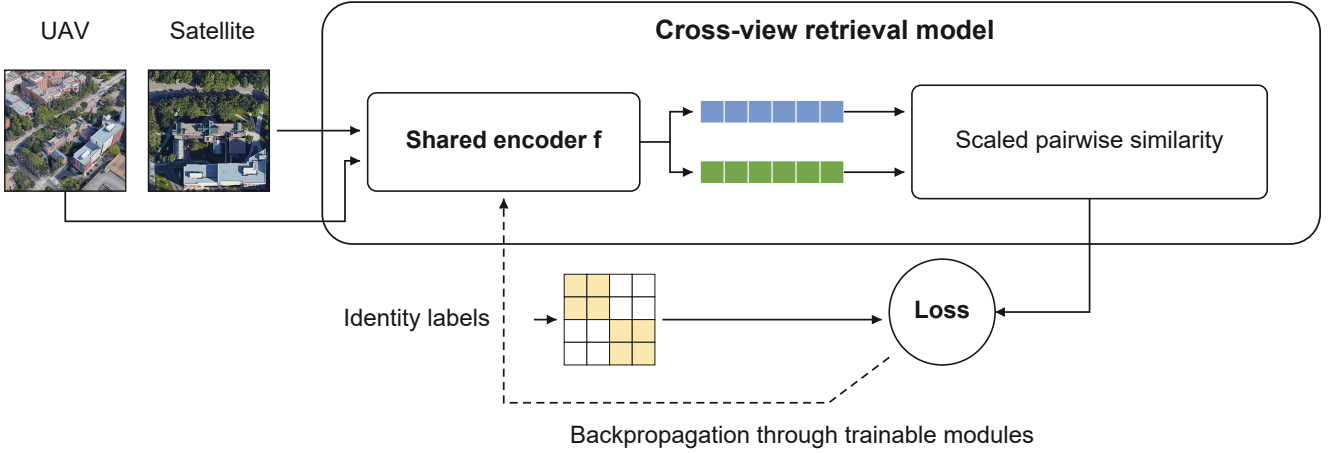
B. Reference Methods and Published Settings

Table II summarizes author-reported results with the corresponding input resources and evaluation setting. MEAN and the MobileGeo core use RGB retrieval inputs. CDM-Net introduces textual descriptions and synthesized views, and MGS2-Net uses a depth prior. MobileGeo’s multi-view postprocessing is listed separately from its core encoder. The two VLGeo rows use the UAV-VLL extension with RGB-only and RGB-plus-text inputs. Grouping these settings preserves the distinction between encoder performance and additional resources available to the retrieval pipeline. The supplementary material reports flight-height comparisons on SUES-200 and the University-1652-to-SUES-200 transfer setting.

C. Training and Model Comparisons

The main configuration uses ResNet-18 [32], a 512-dimensional descriptor, dropout 0.20 [33], resizing to 256 pixels and cropping to 224 pixels. Training is configured for 80 epochs with AdamW [34], base learning rate 3×10^{-4} , backbone learning-rate multiplier 0.1, weight decay 10^{-4} , gradient clipping at 5, and mixed precision [35]. Five warm-up epochs precede cosine annealing without restarts, using the cosine form in [36]. The predetermined final epoch is evaluated without early stopping or test-based checkpoint selection. In this no-validation protocol, the training code names the final completed checkpoint `best.pt`; the filename does not denote selection by validation or test performance. Every training query is scheduled at least once per epoch; the identity sampler and batch construction are specified in Section III.

The main matrix comprises six input configurations on two datasets with three seeds, giving 36 completed fits. Six further

(a) Joint optimization

Frozen CLIP provides image evidence; its weights remain fixed during retrieval training.

(b) Official evaluation tasks

University-1652			SUES-200		
701 training / 701 held-out identities			120 training / 80 held-out identities		
Task	Query	Gallery	Task	Query	Gallery
Drone → satellite	37,855	951	UAV → satellite	4,000	200
Satellite → drone	701	51,355	Satellite → UAV	80	10,000
Street → satellite	2,579	951	Per height: 150 / 200 / 250 / 300 m		

Fig. 4. Joint optimization and official retrieval tasks. (a) UAV and satellite images are independently encoded using shared parameters. Scaled descriptor similarities and the same-identity mask define the multi-positive loss; gradients update the visual backbone and projection/fusion modules while CLIP remains frozen. (b) Official query and gallery counts. University-1652 has 701 training identities, 701 held-out identities, and 250 additional distractor identities. SUES-200 uses the fixed 120/80 training/test identity split and evaluates each of four UAV heights separately; each gallery retains all 200 identities.

seed-1 full-model fits vary the backbone to ResNet-50 or the descriptor width to 256 or 1024 on each dataset. The single-source models test whether a vocabulary alone carries location information; the partial and full models test its incremental effect. QDFL, FSRA, SDPL, CCR, and MCCG remain external visual-comparison targets [7], [9], [16], [17], [19]; this working draft does not report those pending comparisons as completed results. The supplementary material specifies the fit matrix, sampler counts, and reproducibility records.

D. Metrics and Statistical Analysis

The protocol specifies Recall@1/5/10/20 and the public University-1652 evaluator’s trapezoidal mean average precision, denoted mAP_{trap} . Mean reciprocal rank (MRR), per-query rank, top-1 margin, parameter count, peak memory, encoding time, and ranking time provide secondary diagnostics. The reporting unit for the three-seed comparisons is the task-level mean and sample standard deviation. The integrated semantic main and transfer tables are descriptive and do not infer significance from SD. The query diagnostic audit did not

independently resample the retained bootstrap intervals, which are not used here as new inferential evidence.

For the separate completed public-baseline comparisons, 95% intervals use 10,000 query-paired bootstrap resamples unless stated otherwise [37]. Top-1 correctness is compared using two-sided exact McNemar tests [38], [39], with Holm correction [40] within each declared family: 12 directional method pairs for exact reproduction and 32 method–direction–rule comparisons for ranking sensitivity. Comparisons require identical query paths and labels. Because D2S contains repeated queries per location, the selected-fusion analysis also reports identity-cluster bootstrap sensitivity. That analysis treats location as the resampling unit. The selected subset and its evaluation queries are held fixed within these resamples. The supplementary material details the estimands and implementation.

E. Corruption Protocol

The robustness protocol holds galleries clean and corrupts every official query image before re-encoding it. Gaussian

TABLE II
AUTHOR-REPORTED RETRIEVAL RESULTS (%) ON UNIVERSITY-1652 AND ITS UAV-VLL EXTENSION, GROUPED BY INPUT AND EVALUATION SETTING.

Method	Input / protocol	Source	UAV → satellite		Satellite → UAV	
			R@1	AP	R@1	AP
MEAN	RGB, 384	[18]	93.55	94.53	96.01	92.08
MobileGeo (core)	RGB, 224; distillation	[21]	93.87	94.83	95.72	92.57
CDM-Net	Text / generated view, 512	[11]	95.13	96.04	96.68	94.05
MGS2-Net (v2)	Depth prior, 336	[22]	97.60	98.03	98.86	97.25
MobileGeo (+MSRM)	Multi-view postprocessing	[21]	97.15	97.50	95.58	96.27
VLGeo (RGB only)	UAV-VLL RGB, 224	[12]	93.15	96.03	92.01	95.25
VLGeo (RGB + text)	UAV-VLL RGB + text, 224	[12]	98.57	99.17	98.14	98.98

Input size is in pixels. Horizontal groups separate RGB encoders, auxiliary inputs, multi-view postprocessing, and the UAV-VLL extension. Sources: MEAN Table I; CDM-Net Table III; MGS2-Net v2 Table 1; MobileGeo v3 Table I; VLGeo Table II.

noise, Gaussian blur, reduced brightness, and reduced contrast are adaptations of ImageNet-C corruption families [41]. The implemented brightness perturbation darkens images; these family names do not imply identical transformations or severity parameters to ImageNet-C. Center occlusion and rotation are author-defined transformations. Each has five fixed severities. The protocol pairs the visual and full seed-1 models with their clean-reference evaluations. Full uses cached CLIP evidence for clean queries and online CLIP evidence for corrupted queries, so its observed change combines pixel perturbation with a change in evidence path. The 64-image clean diagnostic had no numerical equivalence gate. Results are seed-1 descriptive comparisons, not multi-seed robustness estimates.

V. SEMANTIC-MODEL RESULTS

The completed fixed-protocol evaluation covers 42 completed training runs and 231 run–task evaluations: 36 main fits across six input configurations, two datasets and three seeds, plus six seed-1 sensitivity fits. The main comparisons retain all 11 official tasks, including Street2S and each SUES-200 height and retrieval direction. Task-level summaries weight seeds 1, 2 and 3 equally and report their sample standard deviation (SD), not a confidence interval. No overall score pools tasks, directions or heights.

A. Official Retrieval Across Input Configurations

Tables III and IV report all six configurations. Full has lower mean R@1 and official trapezoidal mAP than Visual on 10 of the 11 tasks. For University-1652 D2S, mean R@1 is 32.300% for Full and 37.864% for Visual; for S2D, it is 44.175% and 54.303%, respectively. All eight SUES-200 tasks have the same direction of difference for both metrics. Street2S is the exception: Full reaches 0.969% R@1 and 2.267% mAP, compared with 0.788% and 1.906% for Visual. Its low absolute performance remains material to the interpretation. These descriptive comparisons do not support a general retrieval-accuracy advantage for the frozen Full configuration.

The content-only and style-only columns retain the standalone vocabulary results, while V+C and V+S retain both

partial combinations. The complete matrix permits comparison of the observed input configurations without selecting favorable tasks or seeds. The additional ResNet-50 and 256/1024-dimensional Full settings are reported with the corresponding 512-dimensional seed-1 reference in the supplementary material. Each sensitivity setting has one fit per dataset; these are not three-seed estimates or capacity-matched causal tests of semantic information.

B. Cross-Dataset Transfer

The transfer evaluation reuses Visual and Full from three source-training seeds in both dataset directions, giving 12 source-model evaluations and 66 seed–task records. University-1652-trained models are evaluated on the eight SUES-200 target tasks; SUES-200-trained models are evaluated on the three University-1652 tasks. Training-dataset direction is distinct from the target task’s query-to-gallery direction.

Table V retains all 22 task–variant groups and their R@1, official mAP and MRR summaries. Full has lower mean R@1 in all 11 target tasks. Mean mAP and MRR are also lower except for the SUES-200-to-University-1652 Street2S task, where the saved Full-minus-Visual differences are 0.040 percentage points for mAP and 0.105 scaled MRR points. Full Street2S R@1 is 0.233% with an actual sample SD of zero, compared with 0.258% for Visual. The small positive rank-sensitive differences do not establish a transfer advantage or statistical significance.

C. Query Corruption and Conditional Diagnostics

The corruption block evaluates Visual and Full at seed 1 under six transformation families and five severities on all 11 tasks, with clean galleries. It contains 660 corrupted task evaluations and 22 clean task–variant references. The retained-percentage curves use $100 m_{\text{corrupt}}/m_{\text{clean}}$ with each variant’s own clean metric. Higher retention therefore does not imply higher absolute corrupted accuracy. Values above 100% remain valid observations, and the analysis is descriptive without multi-seed SD or significance claims. For Full, clean evaluation uses cached CLIP evidence whereas corrupted queries use

TABLE III

OFFICIAL R@1 (%) FOR ALL SIX INPUT CONFIGURATIONS. CELLS SHOW EQUALLY WEIGHTED SEED-1/2/3 MEAN $\pm$ SAMPLE SD. V+C AND V+S DENOTE VISUAL-PLUS-CONTENT AND VISUAL-PLUS-STYLE. SD IS DESCRIPTIVE, NOT A CONFIDENCE INTERVAL. VALUES ARE ROUNDED TO THREE DECIMALS; A POSITIVE VALUE THAT WOULD ROUND TO ZERO IS SHOWN AS < 0.001 . AN ACTUAL ZERO REMAINS 0.000.

Task	Visual	Content	Style	V+C	V+S	Full
University D2S	37.864 $\pm$ 1.011	0.822 $\pm$ 0.012	0.475 $\pm$ 0.004	32.549 $\pm$ 0.245	31.278 $\pm$ 0.203	32.300 $\pm$ 0.347
University S2D	54.303 $\pm$ 3.425	1.522 $\pm$ 0.218	0.095 $\pm$ 0.082	45.364 $\pm$ 0.622	44.223 $\pm$ 2.487	44.175 $\pm$ 1.539
University Street2S	0.788 $\pm$ 0.234	0.129 $\pm$ 0.022	0.026 $\pm$ 0.022	1.112 $\pm$ 0.281	1.112 $\pm$ 0.090	0.969 $\pm$ 0.039
SUES U150 $\rightarrow$ S	35.558 $\pm$ 2.766	1.758 $\pm$ 0.029	0.175 $\pm$ 0.025	28.017 $\pm$ 3.838	27.067 $\pm$ 0.526	28.433 $\pm$ 1.468
SUES S $\rightarrow$ U150	57.500 $\pm$ 3.750	4.583 $\pm$ 0.722	0.000 $\pm$ 0.000	44.167 $\pm$ 5.637	46.667 $\pm$ 1.909	47.917 $\pm$ 0.722
SUES U200 $\rightarrow$ S	42.975 $\pm$ 1.664	2.417 $\pm$ 0.038	0.117 $\pm$ 0.029	34.083 $\pm$ 4.280	35.417 $\pm$ 0.581	33.625 $\pm$ 1.210
SUES S $\rightarrow$ U200	63.333 $\pm$ 2.602	6.250 $\pm$ 1.250	0.833 $\pm$ 0.722	50.417 $\pm$ 4.732	49.167 $\pm$ 4.018	55.000 $\pm$ 5.449
SUES U250 $\rightarrow$ S	46.958 $\pm$ 1.363	2.875 $\pm$ 0.087	0.400 $\pm$ 0.066	37.642 $\pm$ 4.525	37.908 $\pm$ 2.225	36.692 $\pm$ 1.646
SUES S $\rightarrow$ U250	69.167 $\pm$ 3.146	5.833 $\pm$ 0.722	2.083 $\pm$ 0.722	52.500 $\pm$ 4.507	55.833 $\pm$ 2.602	57.083 $\pm$ 4.018
SUES U300 $\rightarrow$ S	49.708 $\pm$ 0.326	3.592 $\pm$ 0.290	0.475 $\pm$ 0.050	39.967 $\pm$ 4.194	39.183 $\pm$ 1.830	38.433 $\pm$ 0.818
SUES S $\rightarrow$ U300	73.750 $\pm$ 3.750	9.583 $\pm$ 0.722	0.000 $\pm$ 0.000	58.333 $\pm$ 2.602	58.750 $\pm$ 0.000	58.333 $\pm$ 1.443

TABLE IV

OFFICIAL TRAPEZOIDAL MAP (%) FOR ALL SIX INPUT CONFIGURATIONS. CELLS SHOW EQUALLY WEIGHTED SEED-1/2/3 MEAN $\pm$ SAMPLE SD. V+C AND V+S DENOTE VISUAL-PLUS-CONTENT AND VISUAL-PLUS-STYLE. SD IS DESCRIPTIVE, NOT A CONFIDENCE INTERVAL. VALUES ARE ROUNDED TO THREE DECIMALS; A POSITIVE VALUE THAT WOULD ROUND TO ZERO IS SHOWN AS < 0.001 . AN ACTUAL ZERO REMAINS 0.000.

Task	Visual	Content	Style	V+C	V+S	Full
University D2S	43.653 $\pm$ 0.950	1.788 $\pm$ 0.015	1.155 $\pm$ 0.009	38.621 $\pm$ 0.288	37.246 $\pm$ 0.314	38.448 $\pm$ 0.269
University S2D	35.620 $\pm$ 1.324	0.740 $\pm$ 0.005	0.326 $\pm$ 0.004	29.100 $\pm$ 0.409	28.551 $\pm$ 0.606	28.970 $\pm$ 0.270
University Street2S	1.906 $\pm$ 0.208	0.670 $\pm$ 0.020	0.432 $\pm$ 0.030	2.324 $\pm$ 0.230	2.273 $\pm$ 0.127	2.267 $\pm$ 0.110
SUES U150 $\rightarrow$ S	43.900 $\pm$ 2.627	5.280 $\pm$ 0.031	1.696 $\pm$ 0.046	36.685 $\pm$ 3.679	35.353 $\pm$ 0.405	36.634 $\pm$ 1.210
SUES S $\rightarrow$ U150	39.523 $\pm$ 2.134	4.051 $\pm$ 0.020	1.439 $\pm$ 0.050	30.765 $\pm$ 1.844	30.890 $\pm$ 0.383	33.178 $\pm$ 1.148
SUES U200 $\rightarrow$ S	51.074 $\pm$ 1.310	6.072 $\pm$ 0.035	1.670 $\pm$ 0.076	42.450 $\pm$ 4.112	42.930 $\pm$ 0.494	41.635 $\pm$ 0.787
SUES S $\rightarrow$ U200	48.755 $\pm$ 1.197	4.267 $\pm$ 0.068	1.441 $\pm$ 0.079	36.588 $\pm$ 3.132	37.785 $\pm$ 0.834	38.964 $\pm$ 1.137
SUES U250 $\rightarrow$ S	54.655 $\pm$ 1.479	6.702 $\pm$ 0.055	1.980 $\pm$ 0.091	45.651 $\pm$ 4.233	45.464 $\pm$ 1.906	44.590 $\pm$ 1.590
SUES S $\rightarrow$ U250	52.298 $\pm$ 1.569	3.722 $\pm$ 0.107	1.557 $\pm$ 0.078	40.865 $\pm$ 2.895	40.467 $\pm$ 0.873	42.746 $\pm$ 1.360
SUES U300 $\rightarrow$ S	57.203 $\pm$ 0.622	7.884 $\pm$ 0.268	2.129 $\pm$ 0.040	47.787 $\pm$ 4.142	46.911 $\pm$ 1.705	46.412 $\pm$ 1.279
SUES S $\rightarrow$ U300	55.070 $\pm$ 0.369	4.423 $\pm$ 0.132	1.563 $\pm$ 0.041	44.238 $\pm$ 4.189	42.523 $\pm$ 1.206	44.500 $\pm$ 0.652

TABLE V

CROSS-DATASET TRANSFER: ALL TARGET TASKS, WITH VISUAL (V) AND FULL (F) SHOWN SEPARATELY. CELLS ARE THREE-SEED MEAN $\pm$ SAMPLE SD.

R@1 AND OFFICIAL MAP ARE PERCENTAGES; $\text{MRR} \times 100$ IS SCALED RECIPROCAL RANK, NOT ACCURACY PERCENT. SD IS NOT A CONFIDENCE INTERVAL. TRAINING-DATASET DIRECTION IS GIVEN BY THE GROUP HEADINGS. VALUES ARE ROUNDED TO THREE DECIMALS; A POSITIVE VALUE THAT WOULD ROUND TO ZERO IS SHOWN AS < 0.001 . AN ACTUAL ZERO REMAINS 0.000.

Target task	R@1 (%)		mAP (%)		MRR $\times$ 100	
	V	F	V	F	V	F
<i>SUES-200-trained $\rightarrow$ University-1652</i>						
University D2S	11.009 $\pm$ 0.117	8.178 $\pm$ 0.609	14.861 $\pm$ 0.063	11.655 $\pm$ 0.700	18.714 $\pm$ 0.038	15.133 $\pm$ 0.809
University S2D	25.678 $\pm$ 1.753	18.260 $\pm$ 1.244	9.543 $\pm$ 0.467	6.967 $\pm$ 0.438	33.549 $\pm$ 1.667	25.885 $\pm$ 2.271
University Street2S	0.258 $\pm$ 0.098	0.233 $\pm$ 0.000	0.704 $\pm$ 0.112	0.743 $\pm$ 0.045	1.149 $\pm$ 0.142	1.254 $\pm$ 0.091
<i>University-1652-trained $\rightarrow$ SUES-200</i>						
SUES U150 $\rightarrow$ S	19.142 $\pm$ 2.331	14.592 $\pm$ 0.927	24.537 $\pm$ 2.395	19.659 $\pm$ 0.666	29.932 $\pm$ 2.497	24.727 $\pm$ 0.663
SUES S $\rightarrow$ U150	19.583 $\pm$ 4.018	12.917 $\pm$ 1.909	12.802 $\pm$ 1.671	9.085 $\pm$ 0.738	25.403 $\pm$ 3.849	18.488 $\pm$ 2.094
SUES U200 $\rightarrow$ S	23.900 $\pm$ 2.718	16.067 $\pm$ 0.797	29.546 $\pm$ 2.784	21.625 $\pm$ 0.707	35.193 $\pm$ 2.883	27.183 $\pm$ 0.660
SUES S $\rightarrow$ U200	26.250 $\pm$ 5.000	15.417 $\pm$ 1.443	16.867 $\pm$ 3.354	12.097 $\pm$ 0.558	33.665 $\pm$ 5.416	21.401 $\pm$ 1.810
SUES U250 $\rightarrow$ S	27.275 $\pm$ 2.368	18.608 $\pm$ 0.986	32.975 $\pm$ 2.463	24.333 $\pm$ 0.777	38.675 $\pm$ 2.575	30.058 $\pm$ 0.598
SUES S $\rightarrow$ U250	30.000 $\pm$ 3.750	20.417 $\pm$ 2.602	21.294 $\pm$ 2.895	15.254 $\pm$ 0.355	36.850 $\pm$ 3.121	26.658 $\pm$ 3.145
SUES U300 $\rightarrow$ S	29.950 $\pm$ 2.792	19.533 $\pm$ 1.102	35.602 $\pm$ 2.919	25.274 $\pm$ 0.971	41.255 $\pm$ 3.059	31.014 $\pm$ 0.851
SUES S $\rightarrow$ U300	34.583 $\pm$ 5.637	19.583 $\pm$ 1.909	24.482 $\pm$ 2.544	16.477 $\pm$ 0.371	40.242 $\pm$ 5.121	26.014 $\pm$ 0.605

online CLIP evidence; the comparison includes this evidence-path change as well as pixel perturbation. The 64-image clean diagnostic did not impose a numerical equivalence gate.

The clean-query diagnostics address a different question: how retrieval outcomes vary with the margin between the leading two cosine scores. T4 uses the same Visual-defined margin-quartile membership for Visual and Full in each task

and seed. Of its 132 margin strata, the 48 SUES-200 satellite-to-UAV strata contain 20 queries each and are descriptive under the registered minimum of 100 queries. This is a Visual-anchored conditional comparison, not a causal explanation of the overall differences.

T5 instead lets each variant select its own margin-ranked subset. Selective risk is one minus R@1 among the selected

queries; coverage-constrained success divides selected-and-correct queries by the common full query count. Equal selected counts do not imply equal memberships. The saved all-prefix AURC is distinct from an area inferred from ten displayed coverage points. The fixed normalized margin score and its ECE diagnostic do not establish posterior probabilities or fitted calibration. These conditional diagnostics do not replace the official and transfer comparisons above. The supplementary material specifies their denominators and evidence limits.

VI. HISTORICAL VISUAL-BASELINE RETRIEVAL ANALYSIS

This separate historical analysis evaluates four public visual baselines on University-1652, beginning with full-gallery reproduction and then examining ranking rules, gallery size and score combination. Its trained feature sources and retrospective fusion are distinct from the semantic configurations in Section V; their gains are not results of the proposed semantic fusion. Within this block, each comparison uses the same four trained feature sources.

A. Exact Full-Gallery Reproduction

Four locally completed University-1652 training runs form the public baseline block: a global ResNet-50 model trained with Google imagery [2], LPN without Google imagery, LPN [6], and LPN with CA-HRS [8]. Their stored feature matrices were independently re-evaluated with the public evaluator’s exact per-query CUDA float32 operation and reverse NumPy sorting. This operation preserves the original evaluator’s floating-point ordering for near-tied matches. Table VI reports the full-gallery metrics. All aggregate metrics, paired confidence intervals, and McNemar comparisons in this subsection use the same exact per-query evaluator outputs after path alignment.

Figure 5(a,b) compares Recall@1 and AP within D2S and S2D, respectively. Panels (c,d) show the query-paired comparisons against CA-HRS. The exact evaluator showed a large separation between the global CNN and the three LPN variants, whereas the LPN variants themselves were tightly grouped. Relative to Global CNN+Google, LPN+CA-HRS increased Recall@1 by 15.03 percentage points in D2S (95% CI, 14.57–15.49; Holm-adjusted $p < 10^{-300}$) and 7.85 points in S2D (95% CI, 5.28–10.41; $p = 2.86 \times 10^{-8}$). By contrast, its differences from LPN without Google and standard LPN were 0.19–0.71 points, and all four corresponding intervals crossed zero (adjusted $p \geq 0.513$). Official mAP showed the same large global-CNN-to-LPN separation. The three LPN variants occupied a much narrower performance range, with Top-1 differences smaller than their paired uncertainty.

B. Retrieval-Rule Sensitivity

Using the same four frozen public feature sources, we separately compared cosine ranking; cross-domain similarity local scaling (CSLS), adapted from local neighborhood scaling [42]; a reciprocal-neighbor bonus, treated as a distinct controlled

variant rather than full k-reciprocal encoding [43]; a rank-weighted average query-expansion heuristic, conceptually related to classical visual automatic query expansion [44]; and query expansion followed by CSLS. We also evaluated all 15 nonempty equal-weight model subsets after score standardization. These experiments quantify post-processing and score combination for the four fixed visual baselines.

Ranking changes affected identification and ranking depth differently. Figure 6 separates D2S and S2D in the upper and lower rows, with Recall@1 and mAP in the left and right columns. CSLS-k10 increased Recall@1 for every method in both directions by 0.97–3.85 points, and all eight paired changes remained significant after Holm correction. Reciprocal-k10 instead reduced D2S Recall@1 by 0.18–0.39 points for all four baselines, with all four reductions remaining significant. AQE and AQE+CSLS changed Top-1 correctness by at most 0.14 points, yet increased S2D official mAP by 6.86–12.24 points. The latter pattern indicates a redistribution of positive ranks rather than additional rank-1 identifications; these mAP shifts are descriptive.

The supplementary material retains the complete numerical retrieval-rule, gallery-scale, and fusion comparison tables.

C. Effect of Gallery Size

Gallery size produced a large and monotonic difficulty shift. Averaged over the four baselines, D2S Recall@1 declined from 88.03% with 100 candidates to 73.06% with all 951 candidates, while official mAP declined from 90.01% to 76.52%. S2D Recall@1 declined from 94.76% with 1,000 candidates to 85.20% with all 51,355 candidates; its official mAP fell more sharply, from 94.87% to 73.13%. The full-gallery method range in the controlled vectorized-ranking analysis was also substantial: D2S Recall@1/mAP spanned 61.92–76.94%/66.43–80.04%, whereas S2D spanned 79.60–87.45%/62.73–76.79%. The 0.01-point differences from the exact per-query reproduction table arise from the distinct near-tie ranking implementations. The envelopes span the minimum and maximum of the four method values. Figure 7 separates the two directions and metrics and retains each method’s complete trajectory. It therefore shows that full-gallery ranking depth, particularly in S2D, is substantially harder than reduced-gallery identification.

D. Error Complementarity and Fusion Cost

The four cosine baselines made non-identical Top-1 errors on the path-aligned University-1652 queries. Figure 8(a) shows all 16 exact correctness patterns as an UpSet display, and panel (b) quantifies overlap between the error sets. The retained three-model fusion reached 80.85% D2S and 89.16% S2D Recall@1, rescuing 2,275 and 21 CA-HRS failures while harming 795 and 9 previously correct queries. Identity-cluster bootstrap sensitivity analysis (10,000 replicates) gave fusion-minus-CA-HRS differences of 3.91 percentage points (95% CI, 3.13–4.70) for D2S and 1.71 points (95% CI, 0.14–3.28) for S2D. The subset was selected on these evaluation queries, so the intervals condition on that selection and quantify sampling variability within the retained subset.

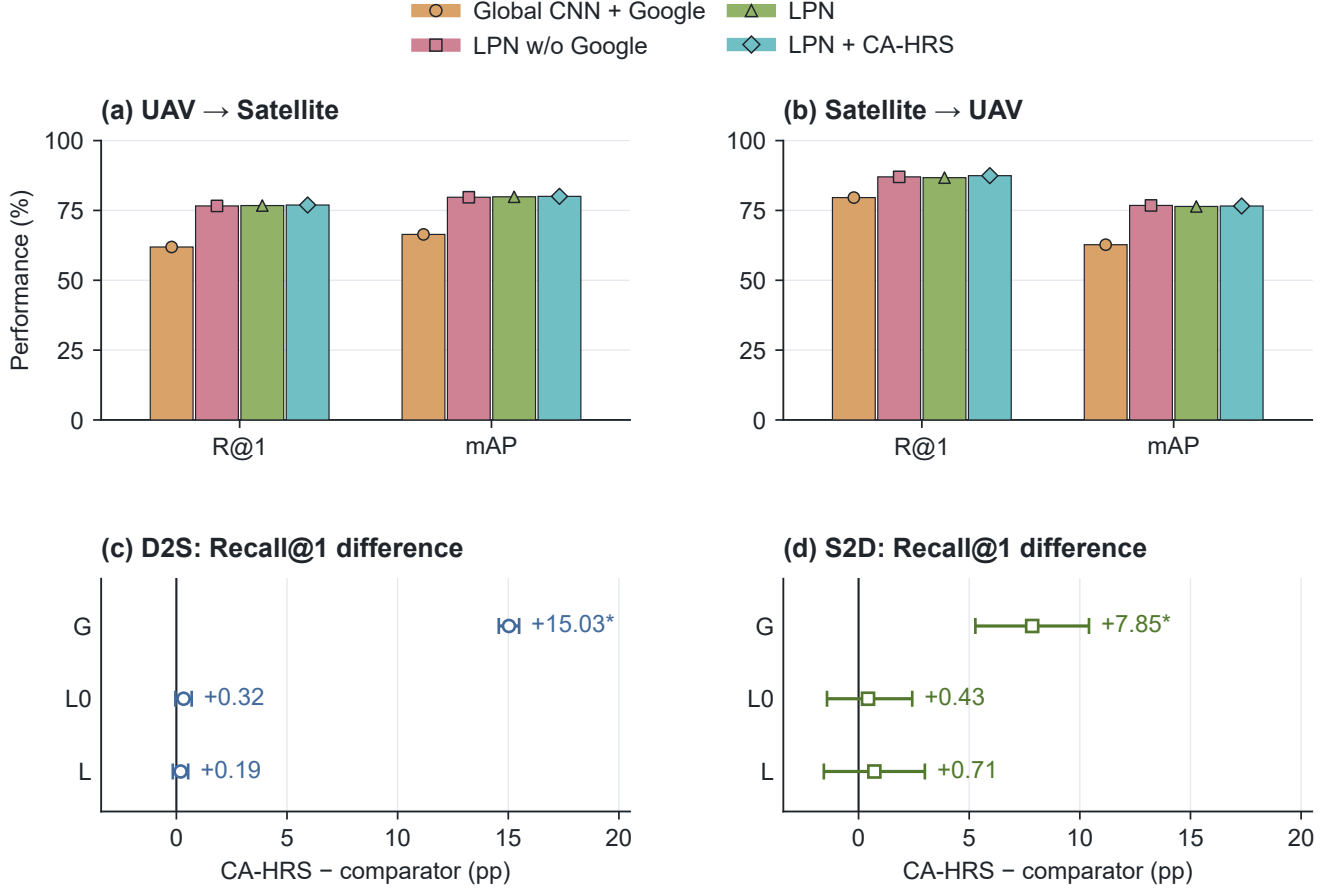


Fig. 5. Exact full-gallery performance and paired effects on University-1652. Grouped bars compare Recall@1 and official trapezoidal mean average precision (mAP) for four public visual baselines in (a) D2S and (b) S2D, using a common 0–100% performance scale. (c,d) CA-HRS-minus-comparator Recall@1 differences for D2S and S2D, with query-paired 95% bootstrap confidence intervals from 10,000 resamples. Asterisks mark exact McNemar comparisons with Holm-adjusted $p < 0.05$ in the retained 12-comparison family. G: Global CNN + Google; L0: LPN without Google; L: LPN; C: LPN + CA-HRS. D2S and S2D denote drone-to-satellite and satellite-to-drone retrieval.

TABLE VI
FULL-GALLERY REPRODUCTION OF PUBLIC VISUAL BASELINES ON UNIVERSITY-1652. ALL VALUES ARE PERCENTAGES. BOLD TYPE INDICATES THE LARGEST VALUE IN EACH COLUMN. AP USES THE OFFICIAL TRAPEZOIDAL EVALUATOR.

Method	Drone → Satellite					Satellite → Drone				
	R@1	R@5	R@10	AP	MRR	R@1	R@5	R@10	AP	MRR
Global CNN + Google	61.91	81.92	87.35	66.42	70.93	79.60	85.59	87.87	62.73	82.39
LPN w/o Google	76.62	90.37	93.60	79.72	82.82	87.02	90.73	92.58	76.79	89.01
LPN	76.75	90.54	93.97	79.88	83.02	86.73	91.30	93.30	76.42	88.97
LPN + CA-HRS	76.94	90.52	93.82	80.04	83.13	87.45	91.87	93.15	76.58	89.51

Figure 9 summarizes the accuracy–cost relation. Table VII gives the corresponding paired comparisons. The selected three-model fusion had higher Top-1 recall and average precision than LPN+CA-HRS on the same evaluation queries. D2S Recall@1/mAP changed from 76.94/80.04% to 80.85/83.52%, whereas S2D changed from 87.45/76.58% to 89.16/80.46%. These gains required three 2,048-dimensional descriptors, with estimated serial retrieval times of 0.52 s (D2S) and 0.67 s (S2D). The estimates add recorded median cosine-ranking time to repeatedly measured fusion overhead. The corresponding

intervals, 0.519–0.532 s and 0.666–0.684 s, reflect overhead variability with cosine time held fixed; they summarize the variability of the measured fusion overhead.

E. Qualitative Retrieval Behavior

Figure 10 shows a Query+Top-5 diagnostic constructed from the frozen public-baseline features. The displayed case is the lexicographically first path-aligned D2S query for which CA-HRS is wrong at Top-1 and the retained fusion is correct; this selection rule is applied before image display. CA-HRS

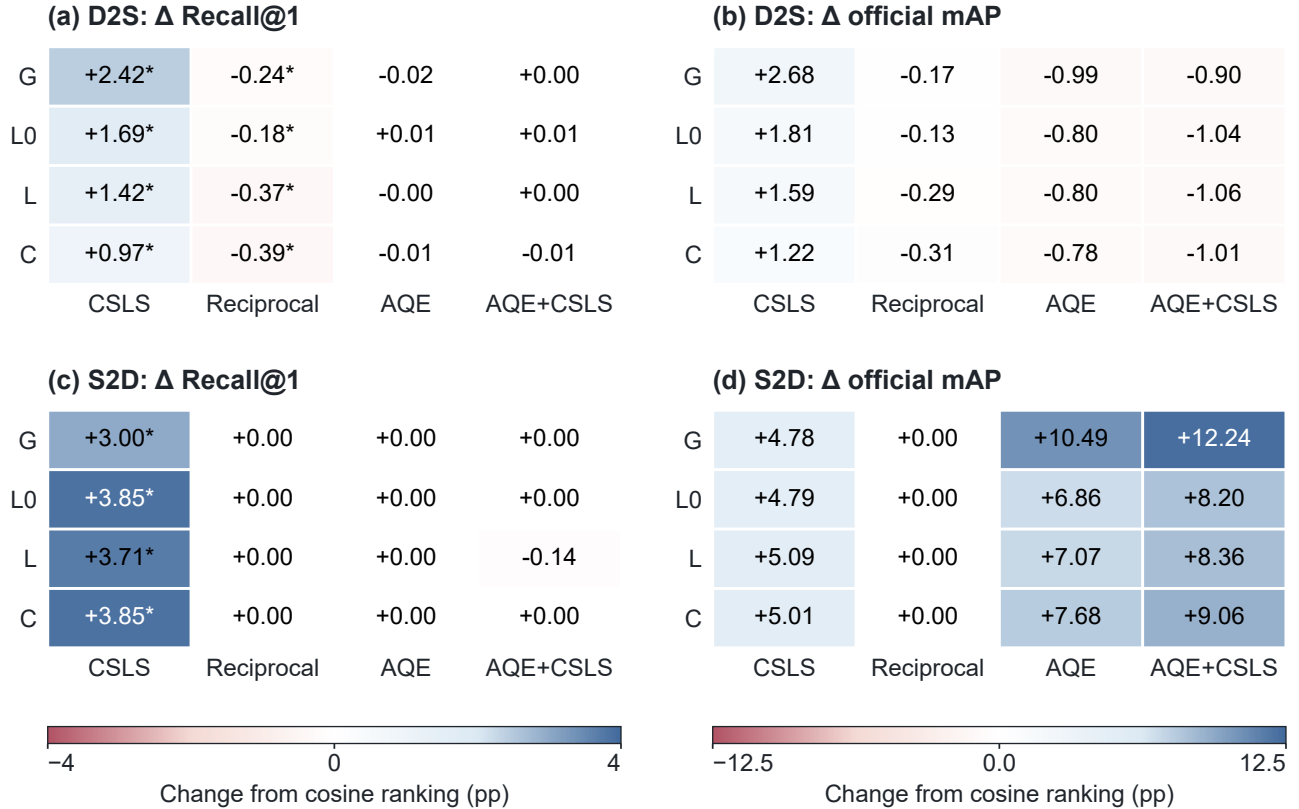


Fig. 6. Controlled retrieval-rule sensitivity using the same frozen public-baseline features. The upper and lower rows show D2S and S2D, respectively. Cells show changes from cosine ranking in (a,c) Recall@1 and (b,d) official trapezoidal mAP, in percentage points. All rules use $k = 10$. Within each metric, both directions share the same zero-centered color scale; the two metrics have different ranges. Blue denotes improvement and red denotes degradation. Asterisks indicate Recall@1 comparisons that pass Holm correction at 0.05 across 32 paired tests; mAP changes are descriptive. Method abbreviations follow Fig. 5.

ranks the correct identity 0005 fourth, behind three visually similar locations; fusion moves identity 0005 to rank 1. The remaining Top-5 candidates contain repeated structures that resemble the query, illustrating the ambiguity among the retrieved gallery images.

VII. DISCUSSION

A. Semantic Inputs and Descriptor Learning

The fixed vocabulary supplies a repeatable input interface, and separate branches preserve the content/style grouping through descriptor construction. The completed comparisons show that this design does not yield a general accuracy advantage under the frozen protocol: Full is lower than Visual in 10 of 11 official tasks for mean R@1 and mAP, and in every transfer task for mean R@1. Street2S provides a small official exception at low absolute accuracy. These are observations about the evaluated configurations; they do not isolate a causal failure mechanism or establish a compensating efficiency, robustness or calibration advantage.

The corruption and query diagnostics need their own interpretation. Retention is normalized by each model’s clean performance, shared Visual margin masks condition the T4 comparison, and T5 variants can select different query members. A favorable conditional value cannot replace the full-task comparison. Independent per-image encoding still permits

gallery descriptors to be computed in advance, but a deployment benefit requires corresponding measurements.

This design also defines concrete modeling trade-offs. Fixed vocabularies provide compact, repeatable inputs but omit detailed object geometry and continuous spatial relations. Their scores depend on the prompt set and the frozen CLIP embedding space; content and style probabilities can remain correlated. Learned probability encoders and descriptor fusion introduce additional parameters. A visual model with comparable capacity is therefore a useful complement to the branch comparisons when attributing an observed effect to semantic information.

B. Ranking, Gallery Composition, and Model Combination

The visual-baseline experiments show that the retrieval procedure affects different parts of the ranking. CSLS changes the leading match across both directions, whereas query expansion mainly improves the placement of additional positives in S2D. Increasing the gallery introduces more competing images and lowers both recall and average precision. These observations support reporting the full gallery, ranking rule, and both metrics alongside the descriptor architecture.

The three-model visual fusion combines partially overlapping error sets. Its rescue and harm counts explain the net improvement over CA-HRS on the analyzed queries. Because

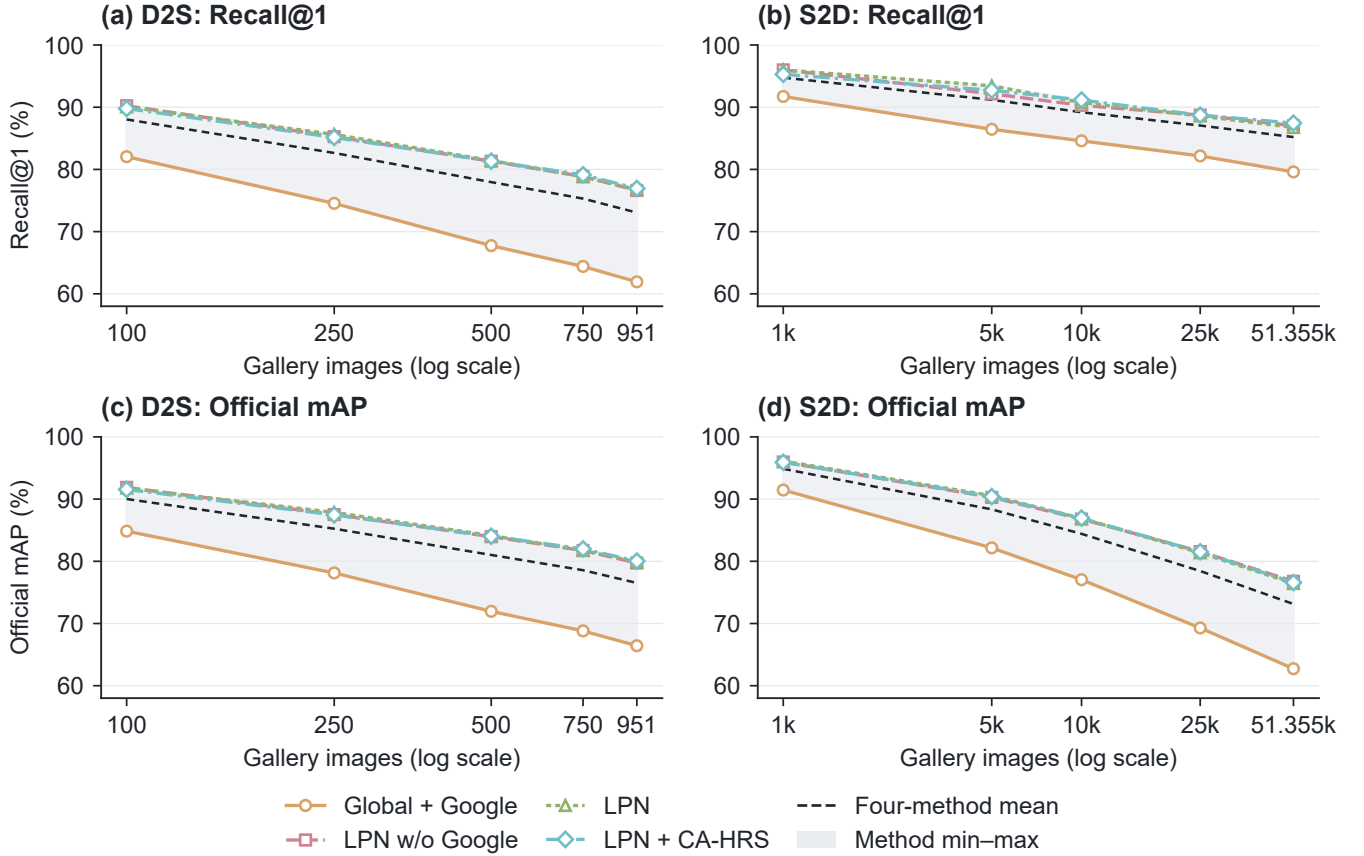


Fig. 7. Gallery-size sensitivity under the controlled vectorized-ranking analysis. (a,b) Recall@1 and (c,d) official trapezoidal mAP for D2S and S2D retrieval, respectively. Colors, marker shapes and line styles identify the four fixed public baselines. Dashed black lines show their arithmetic mean; shading spans the minimum and maximum across methods. Gallery counts use a logarithmic horizontal axis, and all panels share the same performance limits. The near-overlapping LPN trajectories are retained at their original values.

the subset was selected from the same evaluation set, these results describe retrospective complementarity. A deployed ensemble requires a subset fixed on separate validation data. The measured overhead and serial timing estimate also distinguish the cost of combining stored scores from the cost of encoding images with multiple networks.

Location identity is another relevant unit of analysis. University-1652 D2S and Street2S and the SUES-200 UAV-to-satellite tasks contain repeated queries per location. Query-level intervals treat these images as independent observations; identity-cluster resampling retains their shared location membership. The reported fusion analysis includes this cluster analysis, while the broader public-baseline comparisons use query-paired inference. Both the sampling unit and the measured retrieval direction belong in the interpretation of an effect.

C. Extension to Operational Localization

The benchmarks evaluate image identity retrieval across campus and urban scenes. Continuous-region and non-centered localization, as studied in Game4Loc [31], introduces additional spatial uncertainty. Metric pose estimation requires geometric verification after retrieval. Here, the controlled-corruption comparison also changes Full’s evidence path, so

it does not isolate only a pixel transformation. Adverse-weather flights further combine weather, motion, visibility, and capture decisions. Extending the framework to these settings requires data and measurements at the corresponding spatial and environmental scales.

VIII. CONCLUSION

We evaluated fixed content and acquisition-style probability inputs within a shared UAV–satellite retrieval framework. The completed official and transfer matrices retain the full task set and three training seeds. Full has lower mean R@1 and mAP than Visual on 10 of 11 official tasks and lower mean R@1 on all 11 transfer tasks; the low-accuracy Street2S exception does not support a general fusion advantage. Seed-1 corruption and clean-query diagnostics describe relative retention and conditional retrieval under different evidence and query-membership definitions. Together, the results provide a measured assessment of this fixed semantic interface, while the separate public-baseline analysis retains its own ranking and post-hoc fusion conclusions.

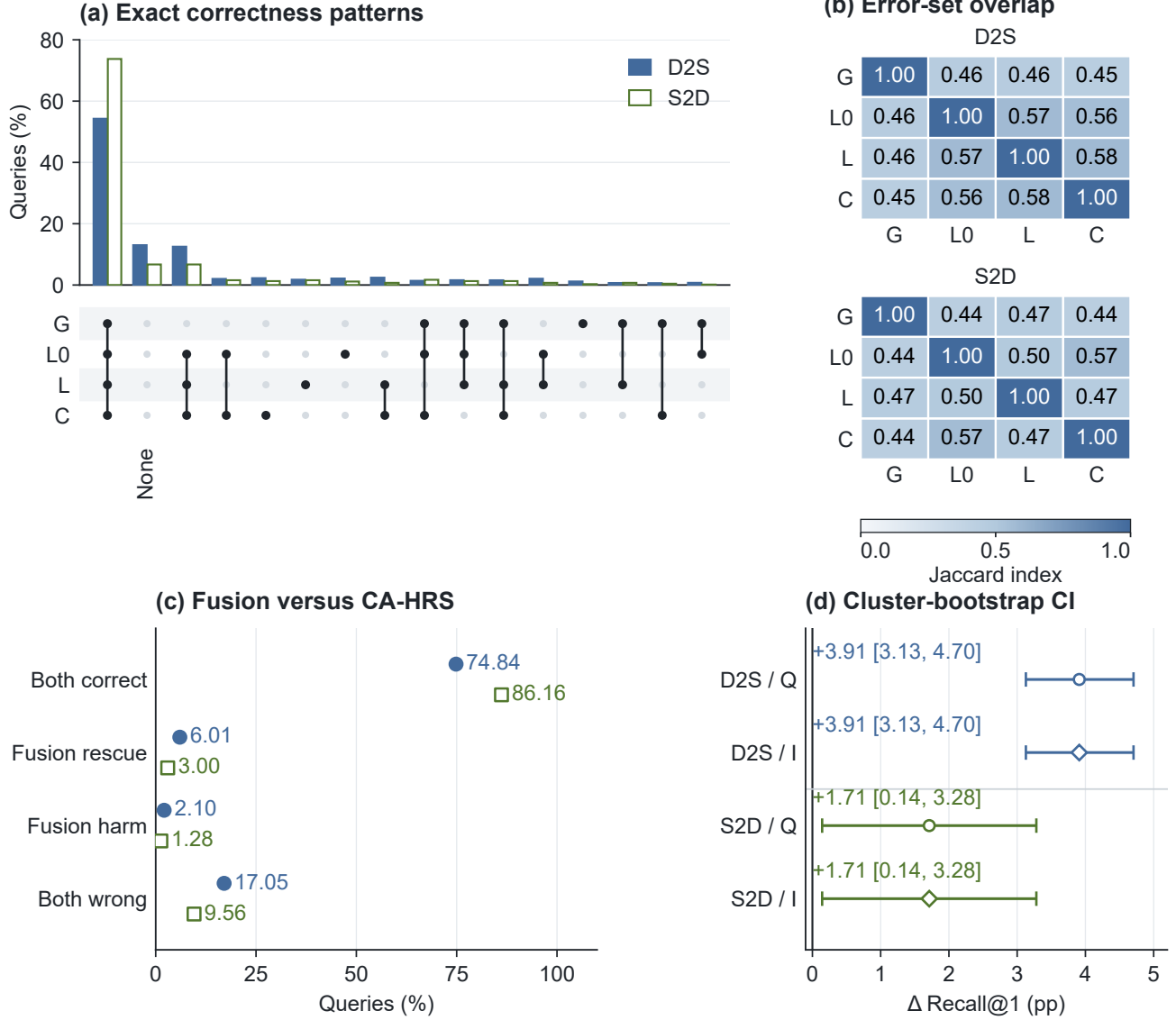


Fig. 8. Query-level complementarity and post-hoc fusion behavior on path-aligned University-1652 queries. (a) All 16 exact Top-1 correctness patterns. Each column of the UpSet matrix specifies which baselines are correct: dark dots indicate inclusion, pale dots indicate absence, and the empty column denotes no correct baseline. Paired bars show the fraction of queries for each direction. (b) Error-set Jaccard overlap, with a common 0–1 scale. (c) The complete partition into both correct, fusion rescue, fusion harm and both wrong relative to CA-HRS. (d) Fusion-minus-CA-HRS Recall@1 differences with 95% identity-cluster bootstrap intervals from 10,000 resamples, under query-weighted (Q) and identity-balanced (I) estimands. The subset was selected on the same evaluation set; intervals are conditional and do not correct selection optimism. Method abbreviations follow Fig. 5.

TABLE VII

POST-HOC THREE-MODEL SCORE FUSION VERSUS LPN+CA-HRS ON PATH-ALIGNED UNIVERSITY-1652 QUERIES. CONFIDENCE INTERVALS USE PAIRED QUERY RESAMPLING; THE TEXT SEPARATELY REPORTS IDENTITY-CLUSTER SENSITIVITY. INTERVALS AND TESTS ARE CONDITIONAL ON THE SELECTED FUSION SUBSET.

Direction	Single R@1	Fusion R@1	$\Delta \text{R@1}$ [95% CI]	Single mAP	Fusion mAP	ΔmAP	p_{Holm}
D2S	76.94	80.85	+3.91 [3.62, 4.19]	80.04	83.52	+3.48	2.71×10^{-163}
S2D	87.45	89.16	+1.71 [0.14, 3.28]	76.58	80.46	+3.88	0.0428

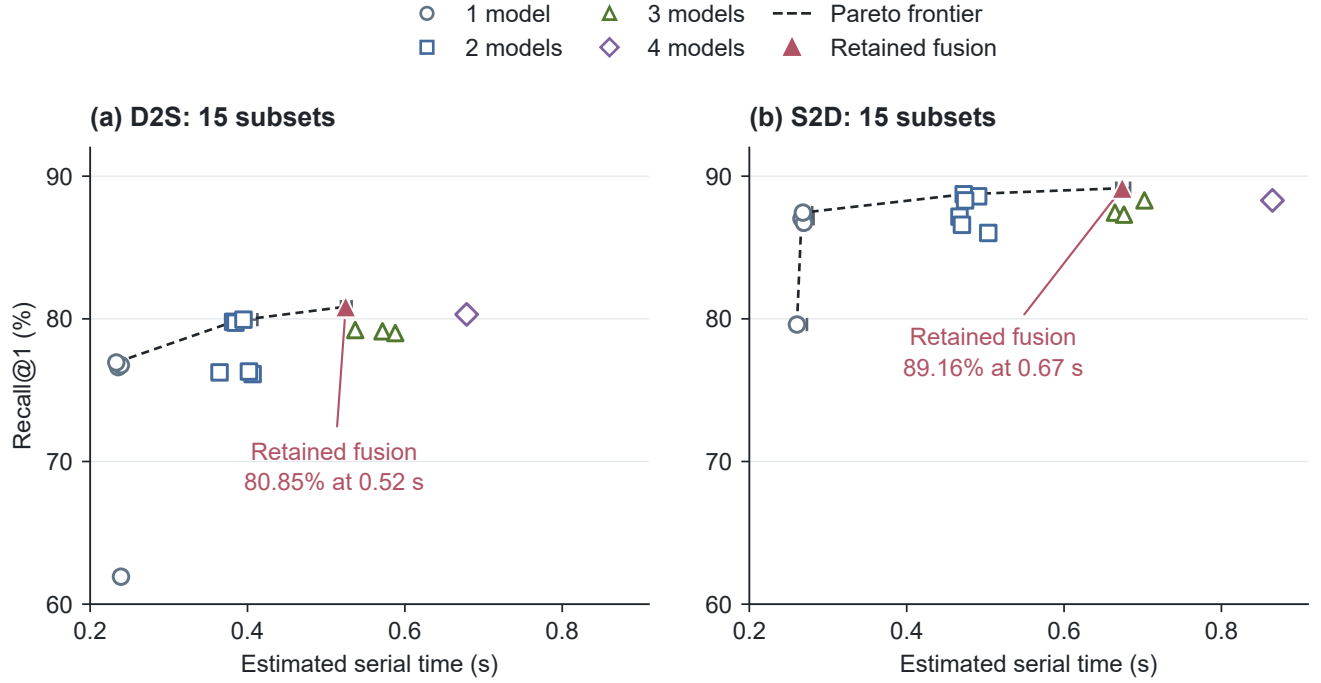


Fig. 9. Accuracy-cost trade-off across all 15 nonempty equal-weight score-fusion subsets of the four public baselines, separately for (a) D2S and (b) S2D. Marker shape and color encode model count. The dashed line joins the Pareto-optimal subsets, and the filled triangle identifies the retained three-model fusion: LPN without Google, LPN and LPN + CA-HRS. Horizontal intervals reproduce the selected overhead Q1–Q3 ranges from 20 timing repetitions with recorded median cosine-ranking time held fixed. They summarize the serial timing proxy obtained from ranking time and fusion overhead. The subset was selected post hoc on the evaluation set.

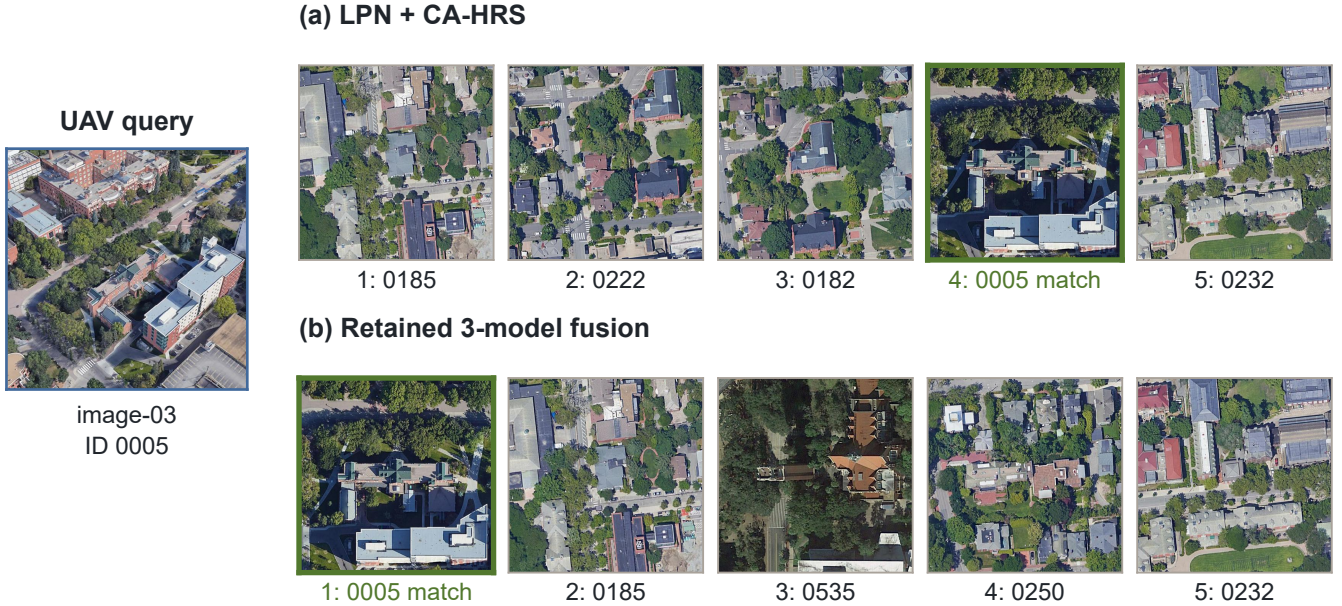


Fig. 10. A deterministic D2S retrieval example from the public-baseline features. The UAV image is the lexicographically first path-aligned query for which CA-HRS fails at Top-1 and the retained score fusion succeeds. Panels compare (a) LPN + CA-HRS and (b) the retained three-model fusion. Labels give rank and gallery identity; green borders and the word “match” identify the correct location. Fusion moves identity 0005 from rank 4 to rank 1. All source images are shown in full, with no cropping, enhancement or image-content modification.

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